

PAPER 2

11:15 am – 12:45 pm (1 hour and 30 minutes)

This paper must be answered in English

INSTRUCTIONS

- (1) This paper consists of **THREE** sections, Section A, Section B and Section C. Attempt **ALL** questions in any **TWO** sections.
- (2) Write your answers in the Answer Book provided. Start each question (not part of a question) on a new page.
- (3) The last page of this Question Paper contains SQL commands and symbols used in entity-relationship diagrams which you may find useful.

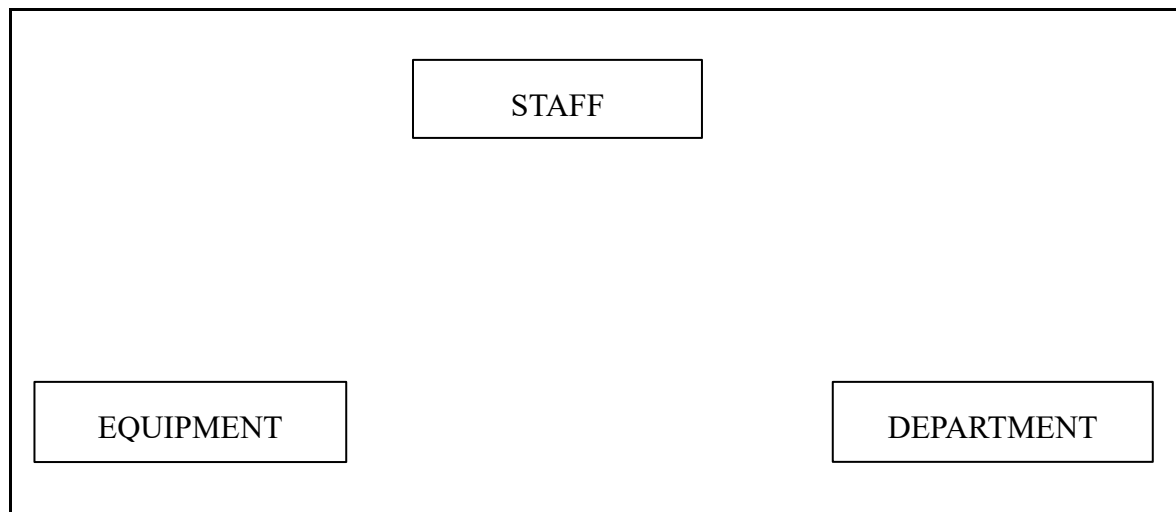
Section A Databases

1. A school uses a database system, “Fix Asset system”, to manage its inventory.
- The equipment will be written off by one staff member. However, a staff member may not write off any equipment.
 - A staff member must belong to a department which has many staff members.

The input screen for the “Fix Asset System - Write off item” is shown below.

Fix Asset System - Write off item	
Write off by	Please select the staff ▼
Department	Information Technology Department
Item no	1625
Description	10.2 iPad 64GB Silver
Date of Purchase	2022-08-04
Write-off Reason	Please Select the following reason ▼
Date of Write-Off Request	Please Select the Date 📅
Request Cancel	

- (a) Design an Entity-Relationship (ER) diagram for database of the school’s inventory management system. It is not necessary to draw attributes. (4 marks)



- (b) Describe how the database management system can handle a lot of errors when multiple operations are executed? (2 marks)

2. The database tables T1 and T2 each have four records with the same structure, as shown below:

T1

CID	AID	AMT
C01	F01	1000
C02	S01	0
C02	S02	6000
C03	F01	0

T2

CID	AID	AMT
C01	F01	1000
C02	S01	500
C02	S02	0
C03	S01	0

- (a) How many records are listed after executing the following SQL statement?

```
SELECT CID, AID FROM T1
UNION ALL
SELECT CID, AID FROM T2
```

(1 mark)

- (b) (i) Complete the following SQL statement to list all CID and AID from T1 that do **not** exist in T2.

```
SELECT T1.CID, T1.AID FROM T1
    (b1) JOIN T2 ON
        T1.CID = T2.CID AND T1.AID = T2.AID
WHERE T2.CID (b2) ;
```

(2 marks)

- (ii) What is the output of the SQL in part (i)?

(1 mark)

3. Ms. Lee uses the database tables STUDENT and MARK to store the student information and their subject marks respectively.

STUDENT

Field Name	Description	Example
SID	Identity code of a student	S2324001
CLASS	Class of a student	2A
FORM	Level of a student	S2

MARK

Field Name	Description	Example
SID	Identity code of a student	S2324001
CID	Identity code of a course	MATH101
SCORE	Score of the course	89

- (a) For the field SCORE of table MARK, integer should be used as the data type instead of text. Explain briefly. (1 mark)
- (b) The following SQL statements are used to complete a task.

SQL statement (S1)	<pre>CREATE VIEW TEMP AS SELECT CLASS, AVG(SCORE) AS AG FROM STUDENT S, MARK M WHERE S.SID = M.SID AND M.CID = "MATH101" AND S.FORM = 'S2' GROUP BY CLASS</pre>
SQL statement (S2)	<pre>SELECT CLASS, AG FROM TEMP WHERE AG = (SELECT MAX(AG) FROM TEMP)</pre>

- (i) State the purpose of S2. (2 marks)
- (ii) State one advantage of using view in the above SQL statements. (1 mark)

4. An airline company uses the database table `FLIGHT` to store flight information. The structure of `FLIGHT` is given below.

`FLIGHT`

Field Name	Description	Example
FID	Flight identity code	AA101
FROM	Flight origin code	HKG
TO	Flight destination code	NRT
FTIME	Flight time	20:00

Sample records of `FLIGHT`

FID	FROM	TO	FTIME
AA101	HKG	NRT	09:00
AA102	HKG	ICN	20:00
AA201	NRT	HKG	07:00
AA202	ICN	HKG	15:45

Flight origin and destination codes in the `FLIGHT` are represented in unique 3-letter codes. Some of the identifier codes are:

HKG: Hong Kong
NRT: Tokyo (Narita)
ICN: Seoul (Incheon)

Airline pilots received a duty roster for the next working day as shown in the following sample:

<u>Flight Roster</u>	
RID: R20240722000017	
Date: 22/07/2024	
Pilot ID: S000017	Name: Peter Chan District: North Asia
Flight ID: AA101 From HKG to NRT Flight Time: 09:00	Flight ID: AA201 From NRT to HKG Flight Time: 20:00

Based on FLIGHT and the information in the roster above, complete the database schema in Third Normal Form below. Identify the corresponding primary keys and foreign keys. Write 'N/A' if not applicable.

LOC (LID, LOCATION)

Primary key: LID

Foreign key: N/A

PILOT (PID, PNAME)

Primary key: PID

Foreign key: N/A

FLIGHT (, , ,)

Primary key: FID

Foreign key: FROM references of LOC

TO references of LOC

ROSTER (, , , ,)

Primary key:

Foreign key: references of FLIGHT

references of PILOT

(5 marks)

5. A public library uses a computer system to manage borrowing records of books. The system uses three database tables MEMBER, LOAN and BOOK to store information on members, books, and loans respectively.

MEMBER

Field Name	Data Type	Width	Description	Example
MID	Character	7	Member identity number	L30245
NAME	Character	30	Name of member	Chan Tai Man
TYPE	Character	1	Member Type: Adult (A) or Junior (J)	A
JOIN_DATE	Date		Date when the member joined	17/5/2023

LOAN

Field Name	Data Type	Width	Description	Example
MID	Character	7	Member identity number	L30245
ISBN	Character	13	International Standard Book Number	9783161484100
LOAN_DATE	Date		Date when the book was borrowed	20/9/2025
STATUS	Character	1	Returned (R) or On loan (L)	R

BOOK

Field Name	Data Type	Width	Description	Example
ISBN	Character	13	International Standard Book Number	9783161484100
TITLE	Character	50	Title of the book	The Great Gatsby
GENRE	Character	15	Genre of the book (e.g., Fiction, Science, History)	Fiction

Write SQL commands to complete the tasks in (a) to (c).

- (a) List the names and joining years of members who are “Junior” members. (2 marks)
- (b) List book titles in the “Science” genre that are on loan. The list should be arranged in ascending order of the book title. (2 marks)
- (c) List the member name and total number of books borrowed by the member with the identity number “L10888”, excluding any books that are in the “Fiction” genre using sub-query. (3 marks)
- (d) Why does the following SQL statement cause an integrity problem?

```
UPDATE LOAN
SET LOAN_DATE = '25/13/2025'
WHERE MID = "L30245" AND ISBN = "9783161484100"
```

(1 mark)

- (e) (i) Briefly describe a way of denormalising MEMBER, LOAN and BOOK. (1 mark)
- (ii) Give **one** advantage and **one** disadvantage of the action in (i). (2 marks)

END OF SECTION A

Section B Web Application Development

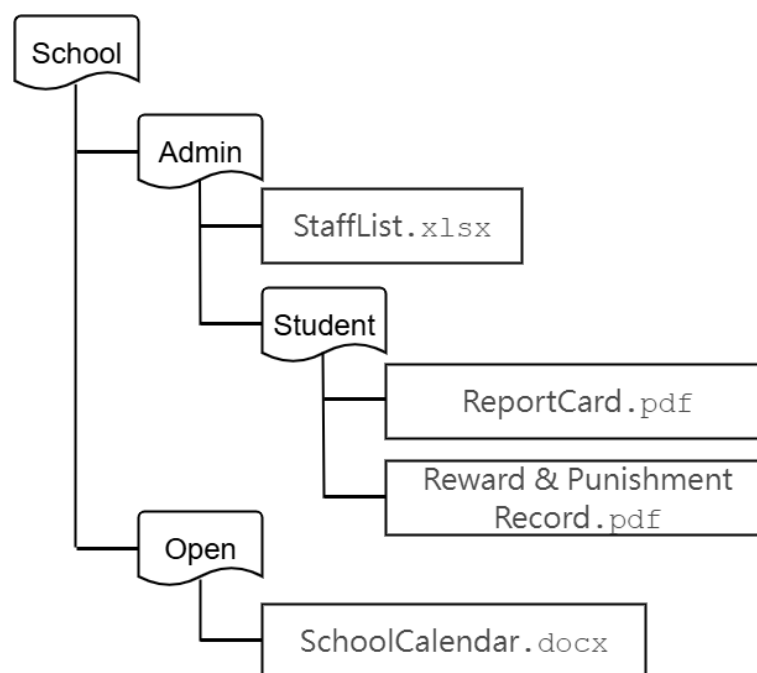
6. A museum ticket reservation system is being developed. The system allows customers to enter the following information on a webpage in order to search for exhibition tickets:

- Visit date (customers can choose any date within the next 90 days)
- Exhibition type (customers should be able to find tickets for different themes using multiple methods)
- Ticket category (Adult, Student, Group)
- Number of tickets (individual purchases are limited to a maximum of 8 tickets per transaction; group purchases must be at least 20 tickets)

(a) As a technical staff member, design an appropriate webpage layout for the system. (3 marks)

(b) Compare hosting your own web server versus using a commercial web hosting service. Provide one advantage and one disadvantage of using a commercial hosting service. (2 marks)

7. The school server's folder structure is as follows:



All permissions granted to a user are inherited by all of that user's sub-folders and files.

- (a) All faculty and staff have "Read & Write" permission on the Admin folder.
Explain one problem that this setting might cause. (1 mark)
- (b) If the Student folder permission is changed to "Read-Only", while the Admin folder permission remains unchanged.
- (i) What effect will this have on faculty members' permissions for ReportCard.pdf? (1 mark)
- (ii) How can you restrict a teacher so that they can only view files in the Student folder for their own class? (1 mark)
- (c) The school needs to share a single folder. Consider two methods:

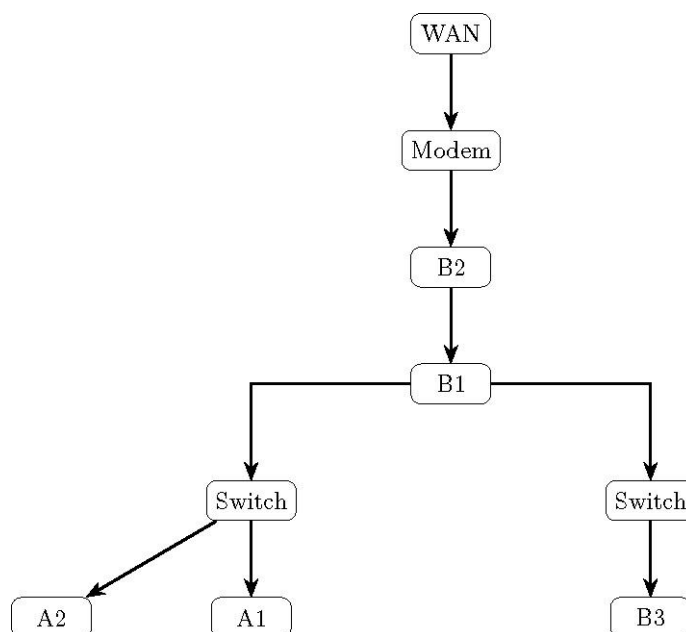
Method 1: Intranet

Method 2: Cloud Storage Service

- (i) State one advantage of Method 1. (1 mark)
- (ii) State one advantage of Method 2. (1 mark)

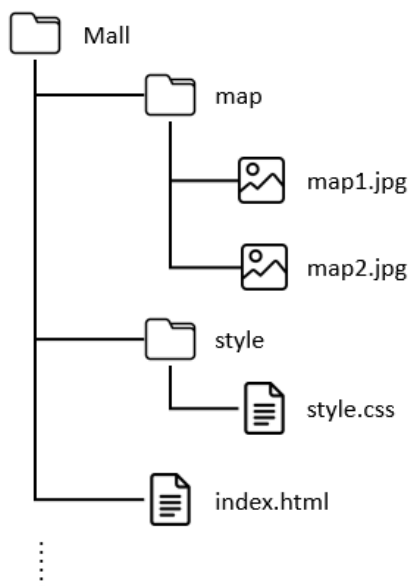
8. The school's network uses the following five distinct devices – A1, A2, B1, B2, and B3 – each of which has a **UNIQUE** function as listed below:

- (1) Authenticate user login credentials
- (2) Cache web content to accelerate access
- (3) Connect the campus wired and wireless networks
- (4) Prevent external unauthorized access to the internal network
- (5) Automatically assign IP addresses to classroom computers



- (a) (i) A1 is a DHCP server. Which numbered function does it correspond to? (1 mark)
- (ii) A2 is a domain controller. Which numbered function does it correspond to? (1 mark)
- (b) Write the name of each device and match it with the corresponding function (from 1 to 5).
- (i) B1 (1 mark)
- (ii) B2 (1 mark)
- (iii) B3 (1 mark)

9. Steve develops a kiosk system that provides an interactive map of the shopping mall. It allows visitors to easily navigate through various stores, food courts, restrooms, and other facilities. Users can also search for specific stores. The structure of some of the folders and files in the server is shown below:



style.css

Line	Code
10	body {
11	font-family: Arial, sans-serif;
12	background-color: #f4f4f4;
13	}
14	.container {
15	max-width: 800px;
16	margin: 20px auto;
17	padding: 20px;
18	background: white;
19	border-radius: 8px;
20	}
21	#map {
22	width: 100%;
23	height: 400px;
24	background: url('Mall/map/map1.jpg') no-repeat;
25	background-size: cover;
26	position: relative;
27	}
28	.search-bar {
29	margin-bottom: 20px;
30	}
31	h5:hover{
32	font-size: 150%;
33	}

index.html

Code

```
<head>
<link rel="stylesheet" type="text/css" href="style/style.css">
.
.
.
<body>
<h1> ACE MALL </h1>
<div class="container">
  <div class="search-bar">
    <input type="text" onkeypress="func()">
  </div>
  <div id="map">
    <div style="padding:200px 15%;">
      <h5>You are here</h5>
    </div>
  </div>
.
.
.
```

- (a) (i) What is the background colour of the webpage? (1 mark)
- (ii) If the mouse cursor of the system is pointing to h5, what will happen? (1 mark)
- (iii) What is the use of onkeypress? (1 mark)
- (b) Steve finds that the image of map1.jpg cannot be loaded. What is the problem? Modify **ONE** line of code in style.css. (2 marks)

The kiosk system is connected to a database in the server. Shop's details are stored in a database table SHOP.

Some sample records are listed below:

NAME	FLOOR	TYPE	LOCATION
More Burgers	2	Restaurant	L2-04
Fine Guys	2	Restaurant	L2-08
Lego Shop	3	Gifts	L3-31
Sushi Go	-1	Restaurant	UG-03

The following HTML code is used to show the searching result in `index.html`.

`shopFinder.php`

Line	Code
101	<code><?php</code>
102	<code>/* The four attributes of mysqli_connect are the hostname,</code>
103	<code>login name, password of the user and database name</code>
104	<code>respectively. */</code>
105	<code>\$conn = mysqli_connect(\$dbhost, \$dbuser, \$dbpass, \$dbname);</code>
106	<code>if(!\$conn){</code>
107	<code> die("Failed to connect!");</code>
108	<code>}</code>
109	<code>\$shop_type = \$_GET['type']</code>
110	<code>\$sql = "SELECT * FROM SHOP WHERE TYPE = '\$shop_type' ";</code>
111	<code>\$result = mysqli_query(\$conn, \$sql);</code>
112	<code>echo "<table>";</code>
113	<code>while(\$row = mysqli_fetch_assoc(\$result)){</code>
114	<code> echo "<tr>";</code>
115	<code> echo "<th> {\$row['NAME']} </th>";</code>
116	<code> echo "<td> {\$row['FLOOR']} </td>";</code>
117	<code> echo "<td> {\$row['TYPE']} </td>";</code>
118	<code> echo "<td> {\$row['LOCATION']} </td>";</code>
119	<code> echo "</tr>";</code>
120	<code>}</code>
121	<code>echo "</table>";</code>
122	<code> Part (d)</code>
123	
124	<code>mysqli_free_result(\$result);</code>
	<code>.</code>
	<code>.</code>
200	<code>.</code>
	<code>?></code>

(c) Steve plans to use cookies to store the current floor of the wayfinding kiosks, so the system can display accurate information to users, such as setting the default floor map and highlighting the shops on that floor.

(i) Briefly describe how to create the cookie `currentFloor` on the kiosk's browser. (1 mark)

- (ii) Complete the missing code below, which will replace line 116, so the system will display different messages if the shop is located on the same floor.

```
If ($row["FLOOR"] == (c ii) ) {  
  
    echo "<td> THIS FLOOR! </td>";  
  
} else {  
  
    echo "<td> {$row['FLOOR']} </td>";  
  
}
```

(1 mark)

- (d) Complete the following code to implement a fixed value check for the shop_type.

```
$accepted = ["Restaurant", "Gifts", "Fashion", "Services"];  
  
if( (d) ) {  
  
    alert("Invalid shop type.");  
  
}
```

(2 marks)

- (e) Complete the missing code below, which will be added in line 123, so the system will display the number of shops found, for example, "3 restaurant(s) found."

```
$count_sql= " (e) ";  
  
$count_result = mysqli_query($conn, $count_sql);  
  
$row = mysqli_fetch_assoc($count_result);  
  
echo $row['total']." ". $shop_type ."(s) found.";
```

(2 marks)

- (f) When a user selects a shop, the system will display a message to help users move to the desired store. The message should indicate whether the shop is

- on the same floor ("The shop is on your current floor."),
- floor above ("The shop is on # floor(s) above."), or
- floor below ("The shop is on # floor(s) below.")

= the floor number difference

shopFinder.php

```
...  
  
<p>Current floor: <strong id="currentFloor">1</strong></p>  
  
<label for="storeSelect">Choose a shop:</label>  
  
<select id="storeSelect"  
  onchange="getGuideMessage(parseInt(document.getElementById('c  
urrentFloor').innerText,10), parseInt(this.value,10))">  
  
  <!-- example options - replace with real data -->  
  
  <option value="1">Store A (floor 1)</option>  
  
  <option value="3">Store B (floor 3)</option>  
  
  <option value="-1">Store C (floor -1)</option>  
  
</select>  
  
<p id="guide"></p>  
  
...
```

Complete the JavaScript function `getGuideMessage` to achieve this.

```
function getGuideMessage (currentFloor, storeFloor) {  
  
  
  
}
```

(3 marks)

- (g) Suggest a reason for including line 124.

(1 mark)

END OF SECTION B

Section C Algorithm and Programming

10. The following algorithm sorts an integer array A with indices from 0 to $N - 1$.

<u>Line</u>	<u>Content</u>
1	for i from 0 to $N - 2$ do
2	for j from 0 to $N - 2$ do
3	if $A[j] > A[j + 1]$ then
4	temp $\leftarrow A[j]$
5	$A[j] \leftarrow A[j + 1]$
6	$A[j + 1] \leftarrow \text{temp}$

Assume $N = 5$ and the initial content of A is

k	0	1	2	3	4
A[k]	13	7	44	16	9

- (a) After executing line 6 for the first time, what are the values of j and A[j]? (1 mark)
- (b) After executing line 6 for the fourth time, what are the values of j and A[j]? (1 mark)
- (c) Name this sorting algorithm. (1 mark)
- (d) Modify line 2 to minimise the number of iterations required for sorting. (1 mark)
- (e) Briefly describe the changes required to sort the array in descending order. (1 mark)

11. Study the following algorithm that takes an integer N as input and uses a queue data structure.

The following subprograms for queue Q are used:

Subprogram	Description
init(Q)	Creates an empty queue Q .
enqueue(Q , X)	Inserts an integer X into the tail of the queue Q .
dequeue(Q)	Removes and returns the head of the queue Q .

Line	Content
1	input N
2	init(Q)
3	temp $\leftarrow N$
4	while temp > 0 do
5	$q \leftarrow$ integer part of (temp / 10)
6	$r \leftarrow$ temp - $q * 10$
7	enqueue(Q , r)
8	temp $\leftarrow q$
9	$A \leftarrow 0$
10	while Q is not empty do
11	digit $\leftarrow$ dequeue(Q)
12	$A \leftarrow A * 10 +$ digit
13	output A

- (a) Dry run the algorithm with the following integer values of input N and provide the corresponding output values.
- (i) 489 (1 mark)
- (ii) 100 (1 mark)
- (b) State the purpose of the algorithm when the input is a positive integer. (1 mark)
- (c) Modify the condition on line 4 so that the algorithm works for both positive and negative input integers. (2 marks)

while _____ (c) _____ do

12. Tim is using a laser cutter. He can input a series of commands to control the engraver at position (x, y) in the diagram below. The initial position of the engraver $(0,0)$ is shown in the diagram below.

$\begin{smallmatrix} y \\ x \end{smallmatrix}$	0	1	2	3	4	...
0						
1						
2						
3						
4						
...						

There are 4 subprograms provided to control the engraver:

Subprogram	Description
GU (d)	Move the engraver d units upward.
GD (d)	Move the engraver d units downward.
GL (d)	Move the engraver d units to the left.
GR (d)	Move the engraver d units to the right.

For example, after executing GD (2) , GR (3) , GU (1) from its initial position, the final position of the engraver will be $(1, 3)$.

$\begin{smallmatrix} y \\ x \end{smallmatrix}$	0	1	2	3	4	...
0						
1						
2						
3						
4						
...						

- (a) After executing GR (1) , GD (2) , GL (1) , GD (1) , GR (2) at its initial position, what is the final position of the engraver? (1 mark)

Tim wants to create a subprogram `GeneratePreview(W, L)` that can store the preview of cutting a $W \times L$ engraved rectangle to a 2D array `Grid`. The preview can be represented as a $(W + 2) \times (L + 2)$ table. All cells on the border are represented by the symbol '#' to denote a cut. The rest of the cells are represented by the symbol '*' to indicate an engraving. `Grid[x, y]` stores the symbol of the cell with position (x, y) .

For example, the diagram below shows the preview of cutting a 3×4 engraved rectangle.

Grid

y x	0	1	2	3	4	5
0	#	#	#	#	#	#
1	#	*	*	*	*	#
2	#	*	*	*	*	#
3	#	*	*	*	*	#
4	#	#	#	#	#	#

- (b) Complete the pseudocode of `GeneratePreview(W, L)`. (4 marks)

```

GeneratePreview(W, L)
  for i from 0 to _____ (b1) do
    for j from 0 to _____ (b2) do

```

(b3)

13. A company, "HKACE Logistics", uses a fleet of drones for package delivery across a city. The city is represented by an $N \times N$ grid of districts.

The company uses the following global variables for its planning program:

Variable	Description
Deliveries	An $N \times N$ integer array. <code>Deliveries[i, j]</code> stores the number of pending deliveries for the district at row i , column j .
N	An integer storing the size of the grid (e.g. if $N = 4$, the grid is 4×4).

In the following example, $N = 4$ and the contents of `Deliveries` are:

$i \backslash j$	0	1	2	3
0	10	5	8	12
1	7	15	3	6
2	2	18	20	4
3	9	1	31	16

- (a) The company wants to find the busiest district in a zone. A zone is a rectangular area defined by its top-left corner $(r1, c1)$ and its bottom-right corner $(r2, c2)$ in the grid.
- (i) Referring to the content of `Deliveries`, what is the number of deliveries in the busiest district, which has the highest number of pending deliveries, within the zone from $(r1, c1) = (1, 1)$ to $(r2, c2) = (2, 3)$? (1 mark)
- (ii) The subprogram `FindBusiest(r1, c1, r2, c2)` returns the number of deliveries in the busiest district, which has the highest number of pending deliveries, within the zone from $(r1, c1)$ to $(r2, c2)$. Complete the pseudocode. (3 marks)

`FindBusiest(r1, c1, r2, c2)`

`m ← 0`

_____ (aii) _____

`return m`

- (iii) Suppose that the values in `Deliveries` could also be negative (representing a net pickup of packages). Explain why the current algorithm for `FindBusiest()` might give an incorrect result. (1 mark)

- (b) To quickly find the total deliveries in any rectangular zone, the company first creates a 2D array called `SumGrid`. `SumGrid[i, j]` stores the total deliveries in the rectangle from (0, 0) to (i, j).

The value of `SumGrid[i, j]` is calculated as:

$$\text{SumGrid}[i, j] = \text{Deliveries}[i, j] + \text{SumGrid}[i - 1, j] + \text{SumGrid}[i, j - 1] - \text{SumGrid}[i - 1, j - 1]$$

- (i) Using the example `Deliveries` grid and the formula above, calculate the value of `SumGrid[1, 1]`. (1 mark)
- (ii) A High-Volume Area is defined as the smallest $K \times K$ square area that starts from (0, 0) and has T or more deliveries. Complete the pseudocode for `FindAreaSize(T)`. It should return K , or -1 if no such area is found. (2 marks)

```
FindAreaSize(T)
  R ← -1
  for K from N downto 1 do
    if _____ (b1) _____ then
      _____ (b2) _____
  return R
```

- (iii) Using `SumGrid`, the total deliveries inside a rectangle from (r1, c1) to (r2, c2) can be with the following formula. Complete the formula below. (3 marks)

$$\text{Total} = \text{SumGrid}[r2, c2] - \text{SumGrid}[\text{_____ (b3) _____}, c2] - \text{SumGrid}[r2, \text{_____ (b4) _____}] + \text{SumGrid}[\text{_____ (b5) _____}]$$

- (c) Managers want to find the k busiest districts. These districts are called hot zones. A 2D array, `isHot`, is used to keep track of those districts. `isHot[i, j]` is set to TRUE(T) if the district (i, j) is one of the k busiest districts. Otherwise, it is set to FALSE(F). Using the example `Deliveries` grid, the following shows the corresponding contents of `isHot` when $k = 5$:

i \ j	0	1	2	3
0	F	F	F	F
1	F	T	F	F
2	F	T	T	F
3	F	F	T	T

The threshold value is the pending deliveries value of k -th busiest district. In the example `Deliveries` grid, the threshold value is 15 when $k = 5$.

- (i) In the example `Deliveries` grid, what is the threshold value when finding the 4 busiest districts (i.e., when $k = 4$)? (1 mark)

- (ii) The company wants to count the connections between hot zones that are next to each other. Using the example Deliveries grid, the number of connections should be 4, as shown in the following diagram when $k = 5$.

$\begin{smallmatrix} j \\ i \end{smallmatrix}$	0	1	2	3
0	F	F	F	F
1	F	T	F	F
2	F	T	T	F
3	F	F	T	T

Diagram showing connections between hot zones (T) in the grid:

- Vertical connection between (1,1) and (2,1)
- Horizontal connection between (2,1) and (2,2)
- Vertical connection between (2,2) and (3,2)
- Horizontal connection between (3,2) and (3,3)

Complete the pseudocode for `CountConnections()`. It checks only neighbours to the right and down. (2 marks)

```

CountConnections()
    c ← 0
    for i from 0 to N - 1 do
        for j from 0 to N - 1 do
            if isHot[i, j] then
                if i < N - 2 and (c1) then
                    c ← c + 1
                if j < N - 2 and (c2) then
                    c ← c + 1
    return c

```

- (iii) In `CountConnections()`, explain why checking only the right and down neighbours is sufficient to count each connection only once. (1 mark)

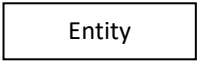
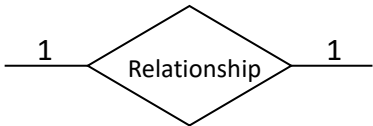
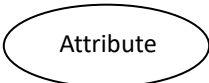
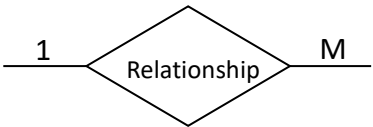
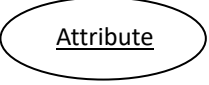
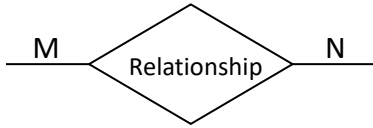
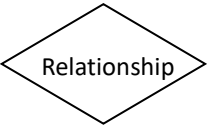
END OF SECTION C

END OF PAPER

Database (SQL commands - based on SQL-92 Standard)

Constants	FALSE, TRUE
Operators	+, -, *, /, >, <, =, >=, <=, <>, %, _ , ' , AND, NOT, OR
SQL	ABSOLUTE (ABS), AVG, INT, MAX, MIN, SUM, COUNT ASC, AT, CHAR (CHR), CHAR_LENGTH (LEN), LOWER, TRIM, SPACE, SUBSTRING (SUBSTR/MID), UPPER, VALUE (VAL) DATE, DAY, MONTH, YEAR ADD, ALL, ALTER, ANY, AS, ASC, BETWEEN, BY, CREATE, DELETE, DESC, DISTINCT, DROP, EXISTS, FROM, GROUP, HAVING, IN, INDEX, INNER JOIN, INSERT, INTEGER, INTERSECT, INTO, LEFT [OUTER] JOIN, LIKE, MINUS, NULL, RIGHT [OUTER] JOIN, FULL [OUTER] JOIN, ON, ORDER, SELECT, SET, TABLE, TO, UNION, UNIQUE, UPDATE, VALUES, VIEW, WHERE

Symbols Used in Entity-Relationship Diagrams

Meaning	Symbol	Meaning	Symbol
Entity		One-to-One Relationship	
Attribute		One-to-Many Relationship	
Key Attribute		Many-to-Many Relationship	
Relationship		Participation constraints: Use  on Mandatory side Use  on Optional side	